

Alexander Hölzemann

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[Github](#) | [LinkedIn](#) | [Google Scholar](#) (h-index: 10) | [Full Academic CV](#)

SKILLS

ML & Modeling: Keras/TensorFlow, PyTorch (basic), scikit-learn, self-supervised, supervised and unsupervised learning

Data & Scientific Computing: NumPy, Pandas, SciPy, xarray, Zarr, Dask

Visualization: Matplotlib, Plotly

Compute & Cloud: HTCondor, HTC/HPC environments, Docker (basic), AWS (S3, boto3)

Experimentation: Weights & Biases, Optuna

Engineering & Collaboration: Git, Jupyter, Slack/Mattermost, LaTeX

Additional / Prior: SQL, Unix/Bash, Java, JavaScript/TypeScript, C/C++, R, Dart, Flutter, Angular,

Languages: German (native), English (C2), Spanish (C1)

PROFESSIONAL EXPERIENCE

2026 **Postdoctoral Researcher | Department of Computer Science | University of Colorado Boulder, Boulder, USA | September 2024 - Present**

- Engineered a self-supervised learning pipeline to separate biological acoustic signatures from background signals and support fish species and schooling-pattern classification.
- Developed Autoencoder/Decoder and BYOL models and implemented multilevel BGMM clustering with sub-clustering to isolate biological signatures from background.
- Architected cloud-native deep learning pipelines to stream NOAA petabyte-scale [water-column sonar data](#) from AWS S3 using Zarr and Xarray, eliminating local storage limits.
- Engineered a scalable machine learning pipeline optimized for distributed execution and high-throughput processing on a parallelized [HTC cluster](#).
- Contributed to the Project Pythia [OSDF-Cookbook](#), developing reproducible Jupyter Notebook workflows for streaming [NOAA Water-Column Sonar Data](#) via the Open Science Data Federation (OSDF) to enable cloud-based, scalable scientific analysis.

2024 **Scientific Researcher and Ph.D. Candidate | Department of Ubiquitous Computing | University of Siegen, Germany | May 2018 – August 2024**

- Developed machine and deep learning pipelines and models for Human Activity Recognition (HAR)
- Investigated transfer learning behavior for wearable-sensor HAR ([Project: Digging Deeper](#)) and implemented end-to-end evaluation pipelines.
- Developed a cross-correlation algorithm to synchronize data streams from multiple wearable IMUs, aligning timestamps with millisecond precision.
- Architected full-stack data collection tooling ([Project: ActiVATE](#)), building a cross-platform mobile application (Flutter) alongside a backend and database to store participant data.
- Led study design and data collection for basketball activity recognition ([Project: Hang-Time HAR](#)), annotating over 50 hours of data and releasing a benchmark dataset via [Zenodo](#).

EDUCATION

- 2024 **Ph.D. in Computer Science**, Department of Ubiquitous Computing,
University of Siegen, 57076 Siegen, Germany, Final Grade: Magna Cum Laude
- 2018 **Master of Science in Computer Science**, University of Siegen, 57076 Siegen, Germany
- 2013 **Bachelor of Science in Computer Science**, University of Siegen, 57076 Siegen, Germany

SELECTED PUBLISHED MANUSCRIPTS

Peer-reviewed journals, conferences or workshops.

Google Scholar h-Index 10, 402 Citations (as of May 2026) - Full list is available on Google Scholar

- 2024 **Hoelzemann, A.**, and Van Laerhoven K.. "A matter of annotation: an empirical study on in situ and self-recall activity annotations from wearable sensors", *Frontiers in Computer Science*, [DOI](#).
- 2023 **Hoelzemann, A.**, et al. "Hang-Time HAR: A Benchmark Dataset for Basketball Activity Recognition Using Wrist-Worn Inertial Sensors", *MDPI Sensors*, [DOI](#).
- 2021 Bock, M., **Hoelzemann, A.**, et al: "Improving Deep Learning for HAR with Shallow LSTMs", *ACM ISWC*, [DOI](#). ☆ **BEST PAPER AWARD**
- 2021 **Hoelzemann, A.**, Sorathiya, N., & Van Laerhoven, K. "Data Augmentation Strategies for Human Activity Data Using Generative Adversarial Neural Networks", *IEEE PerCom*, [DOI](#).
- 2020 **Hoelzemann, A.**, & Van Laerhoven, K. "Digging Deeper: Towards a Better Understanding of Transfer Learning for Human Activity Recognition", *ACM ISWC*, [DOI](#)

LEADERSHIP & MENTORING

Project & Event Leadership: Co-General Chair for international workshops (WellComp [2023/2024](#)), overseeing committee appointments, budget, and on-site operations.

Team Coordination: Recruited and managed international teams of student volunteers for large-scale conferences ([UbiComp/ISWC](#)).

Technical Mentoring: Designed and delivered a [tutorial](#) on Deep Learning for HAR, covering end-to-end sensor data preprocessing, noise filtering, and class balancing for model training.

Education: Guided students in applying deep learning methods to practical projects and supervised scientific writing.

PROFESSIONAL INTERNSHIPS & EARLY EXPERIENCE

- 2018 **Research Assistant** | University of Siegen, Germany | 2014 – 2018
- 2016 **Working Student (Mobile & Web Dev)** | Krombacher Brewery, Germany
- 2015 **IT Internship** | Oiltanking Ebytem S.A., Argentina
- 2015 **Working Student (1st Level Support)** | SMS Siemag AG, Germany | 2010 – 2015